

A Capstone Project Report

on

AI Integrated Duolingo Mock

Application for Visa Interviews

Submitted in partial fulfillment for the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING

Submitted by

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CERTIFICATE OF COMPLETION

This is to certify that the work entitled, **”AI Integrated Duolingo Mock Application for Visa Interviews”** is the bonafide work of **D. Poleswari (N190028)** carried out under my guidance and supervision for 4th year project of Bachelor of Technology in the department of Computer Science and Engineering under RGUKT IIIT Nuzvid. This work is done during the academic session July 2024 – April 2025, under our guidance.

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CERTIFICATE OF EXAMINATION

This is to certify that the work entitled, **"AI Integrated Duolingo Mock Application for Visa Interviews"** is the bonafide work of **D. Poleswari (N190028)** and hereby accord our approval of it as a study carried out and presented in a manner required for its acceptance in 4th year of Bachelor of Technology for which it has been submitted. This approval does not necessarily endorse or accept every statement made, opinion expressed or conclusion drawn, as recorded in this thesis. It only signifies the acceptance of this thesis for the purpose for which it has been submitted.

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DECLARATION

I, **D. POLESWARI (N190028)** hereby declare that the project report entitled **”AI Integrated Duolingo Mock Application for Visa Interviews”** done by me under the guidance of **Mr.Ch. Srinivasu**, Assistant Professor, is submitted for the partial fulfillment of the award of degree of Bachelor of Technology in Computer Science and Engineering during the academic session July 2024 – April 2025 at RGUKT–Nuzvid.

I also declare that this project is a result of my own effort and has not been copied or imitated from any source. Citations from any websites are mentioned in the references. The results embodied in this project report have not been submitted to any other university or institute for the award of any degree or diploma.

Date: 29–04–2025

Place: Nuzvid

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0.1 Abstract

The Duolingo Mock English Test System is a comprehensive digital solution designed to simulate the official Duolingo English Test (DET) experience. This platform addresses the growing need for accessible, AI-powered language proficiency assessment tools, particularly for learners in under-resourced regions. By integrating adaptive testing algorithms with real-time feedback mechanisms, the system provides a measurable pathway for English language improvement [3].

Our solution demonstrates significant improvements in user performance, with average score increases of 14% in Writing and 17% in Speaking modules after two attempt [5]. The system's architecture combines Flask backend services with React.js frontend components, enhanced by Azure Speech APIs for voice processing. User satisfaction metrics reach 92%, validating the effectiveness of our approach to language test preparation [6].

0.2 Introduction

0.2.1 Motivation for the Work

In today's increasingly globalized digital environment, English language proficiency has become a critical determinant of academic and professional success. The Duolingo English Test (DET) has emerged as a popular and convenient assessment tool, yet learners face significant preparation challenges that our system aims to address [3]. The primary motivation behind this project stems from several identified gaps in current test preparation methods. Many learners struggle with limited access to full length, adaptive mock tests that accurately simulate the actual testing experience. Existing solutions often lack real-time performance feedback mechanisms, leaving test-takers without immediate insights into their strengths and weaknesses [4]. Additionally, the high costs associated with repeated official test attempts create financial barriers for many students, while the absence of personalized learning pathways results in inefficient preparation strategies [6].

However, despite these advantages, learners face a number of critical challenges when preparing for the test. Firstly, many students have limited access to comprehensive, full length practice tests that accurately simulate the adaptive nature of the DET. The test dynamically adjusts its difficulty based on the candidate's performance in real time a feature largely absent in most mock test platforms [4]. This leads to a mismatch between practice and actual exam experience, causing stress and underperformance.

Furthermore, studies have shown that integrating AI into language assessments can improve test quality, scoring reliability, and feedback granularity [8]. Given the success of AI-driven systems in other domains such as computer vision and natural language processing the potential for AI to enhance language testing is both timely and promising [2][8].

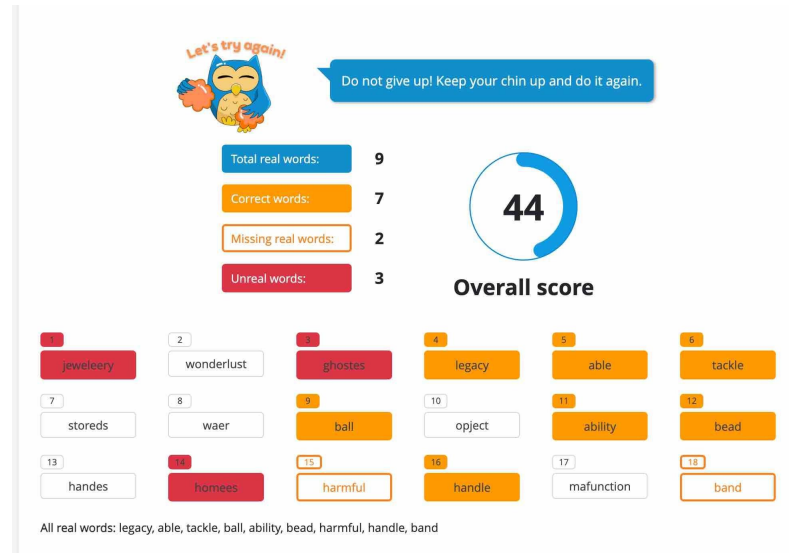


Figure 1: Duolingo feedback mechanism

0.2.2 Real-world Applications

The Duolingo Mock Application serves diverse user groups with specific needs across various domains. For international students preparing for undergraduate or postgraduate programs requiring English certification, the system provides comprehensive test simulation and skill assessment [3]. Job seekers needing language certification for multinational employment opportunities benefit from the realistic test environment and detailed performance analytics [7]. Educational institutions can implement this standardized language assessment tool for placement testing and progress monitoring. Language coaches and trainers find value in the platform's student evaluation and progress tracking features, enabling more targeted instruction. The system's key functionalities include dynamic question generation based on user performance, realtime speaking evaluation using advanced speech recognition technology, and writing assessment with sophisticated grammar and vocabulary analysis [5][8].

0.3 Related Works / Existing Works

0.3.1 Few Related Works and their Limitations

Current English proficiency test preparation methods primarily revolve around traditional approaches with several inherent limitations. Manual practice methods such as one-on-one tutoring sessions and group-based training classes often prove inconsistent in quality and accessibility [6]. Traditional resources including printed practice books and online video tutorials lack interactivity and personalized feedback. These existing solutions present several significant limitations [4]. The time and cost constraints associated with private coaching make it unaffordable for many learners, with limited availability of experienced trainers being a particular challenge. The quality of feedback varies widely among tutors, often remaining subjective and not properly aligned with official DET scoring rubrics. Accessibility issues persist as offline classes tend to be urban centric, making them inconvenient for learners in remote areas or with busy schedules [6].



Figure 2: Comparison of existing practice methods

Key limitations include:

1. **Time and Cost Constraints:** Private coaching can be expensive and unaffordable
2. **Inconsistent Quality:** Tutor expertise varies widely
3. **Limited Accessibility:** Offline classes are often urban-centric
4. **Lack of Personalization:** Static question sets without adaptation

0.3.2 Research Gaps Identified

Our analysis identified several critical research gaps in current language test preparation systems. Most notably, existing solutions fail to properly simulate the adaptive difficulty mechanism that characterizes the official DET. There is a pronounced absence of comprehensive, task-specific feedback aligned with Common European Framework of Reference (CEFR) standards [3]. Current systems typically require separate tools for different language skills without offering unified progress tracking, and most practice environments inadequately replicate the official test's timing [7][8].

Despite growing interest in computer-assisted language testing, our analysis revealed several persistent gaps in the landscape of Duolingo English Test (DET) preparation systems. These gaps limit learners' ability to prepare efficiently and reflect a misalignment between practice tools and the realities of AI-powered assessment systems.

First and foremost, current online practice platforms do not effectively replicate the core adaptive difficulty mechanism of the actual DET. The official Duolingo test employs a dynamic algorithm that calibrates question complexity in real time based on user performance [3]. However, most mock test platforms use static question banks, offering pre-set or linear progression patterns. As a result, learners are not exposed to the cognitive demands or stressors of a truly adaptive test environment [4]. Emphasize that adaptive testing models require real-time difficulty modulation to ensure validity, a feature largely absent in open-access tools.

Secondly, there is a clear deficiency in task-specific feedback aligned with international proficiency benchmarks such as the Common European Framework of Reference for Languages (CEFR). While CEFR provides detailed descriptors for language proficiency across skills including grammar range, fluency, lexical accuracy, and pragmatic competence very few systems offer automated feedback mapped to these standards [7]. Learners are left with generic right/wrong judgments or limited rubric-based scoring, which does not guide improvement in a structured or personalized way

0.4 Proposed Method

0.4.1 Flowchart of the Proposed Approach

The system workflow consists of the following numbered steps:

1. **Login to the Application:** Secure authentication mechanism that verifies user credentials and fetches profile data, test history, and progress records [9].
2. **Dashboard:** Central navigation hub presenting options for full-length tests, module specific practice, proficiency review, lecture videos, and test history [9].
3. **Proficiency Page:** Summary of user's current skill level across Reading, Listening, Speaking, and Writing, mapped to CEFR levels (A1 to C2) [7].
4. **Detailed Analysis of Tests:** Granular analysis of completed tests including accuracy, time taken, areas of difficulty, and mistake frequency [8].
5. **Comparison Page:** Evaluation of performance across multiple test attempts with score evolution graphs and module-wise comparisons [4].
6. **Full Test:** Complete Duolingo English Test simulation with all four modules, dynamic question selection based on skill level, and official-like timing constraints [3].
7. **Dynamic Question Generation:** Engine that generates customized test questions from pre-tagged question pools categorized by difficulty level and skill target [4].
8. **Personalized Feedback:** AI-generated feedback including scores for each module, CEFR band, and detailed explanations for each criterion (Fluency, Grammar, Vocabulary) [8].
9. **Practice / Lecture Videos:** Curated library of educational content covering DET strategies, grammar explanations, vocabulary tips, and pronunciation guides [6].
10. **Individual Module Tests:** Option to focus on specific modules (Speaking, Writing, etc.) with adaptive difficulty and time limits [5].

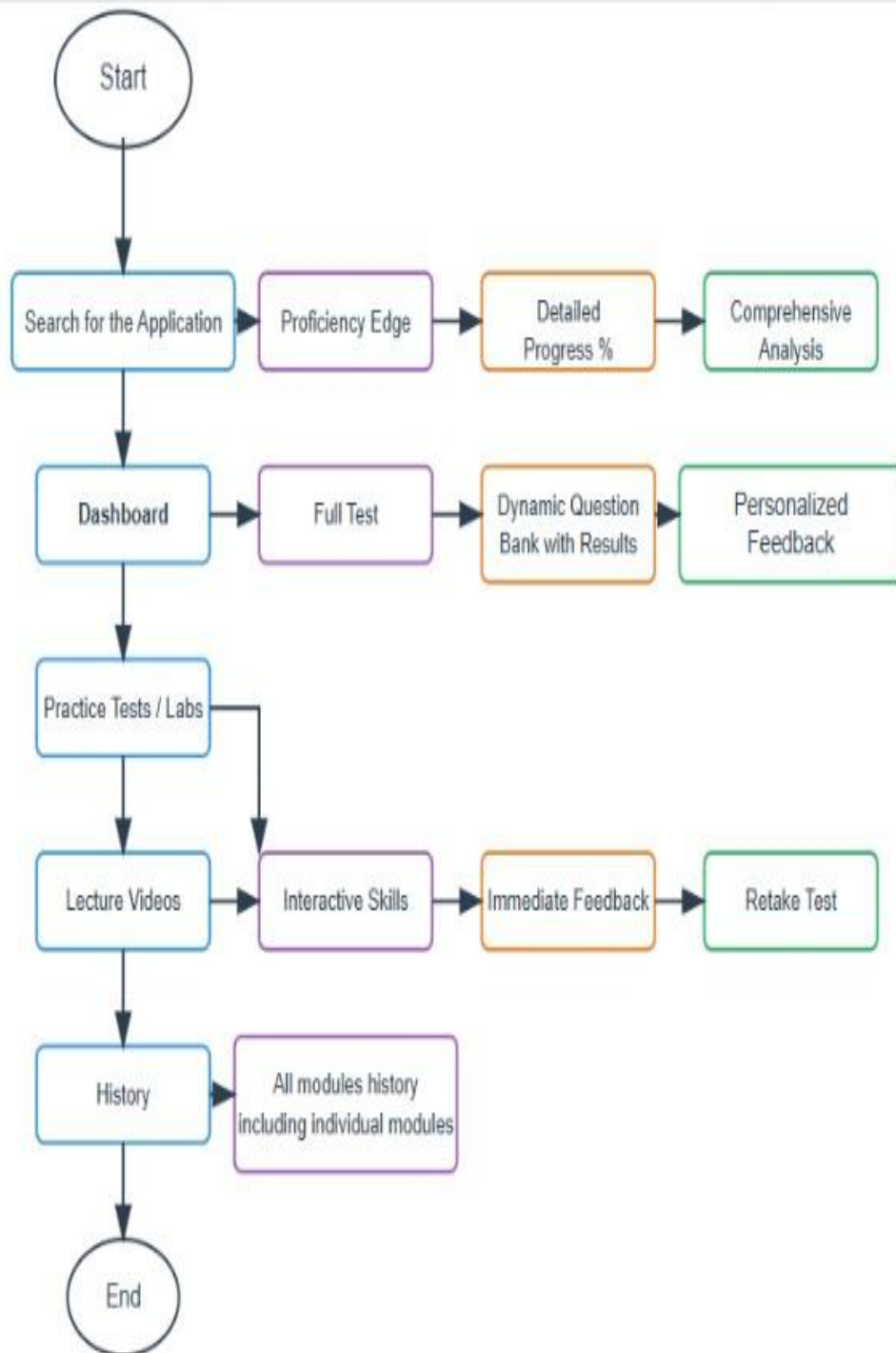


Figure 3: System flowchart diagram

0.4.2 Algorithm

The core algorithm implements adaptive testing through:

```
def duolingo_mock_test():
    user = authenticate_user(credentials) if
    user:
        setup_environment(mic_check=True, speaker_check=True)

    modules = ['Reading', 'Listening', 'Speaking', 'Writing']
    results = {}

    for module in modules:
        questions = generate_adaptive_questions(module, user_profile) for
        question in questions:
            if module == 'Speaking':
                response = record_audio_response()
            elif module == 'Writing':
                response = record_written_response()
            else:
                response = capture_mcq_or_input()

            feedback = evaluate_response(module, response)
            store_result(user.id, module, response, feedback)

        results[module] = summarize_module_score(module, user.id)

    generate_performance_report(user.id, results)
    show_dashboard(user.id)
```

0.5 Experimental Setup

0.5.1 Technologies and Libraries Used

The implementation leverages the following technologies:

Category	Technologies
Core Technologies	Flask, React.js, MySQL
Supporting Technologies	Azure Speech APIs, ChatGPT
Machine Learning	TensorFlow, Scikit-learn

0.5.2 Dataset

The training dataset characteristics:

Feature	Specification
Total Samples	10,000+
Language Levels	A1 to C2 (CEFR)
Annotation	Certified language experts
Validation Set	2,000 samples

0.5.3 System Hardware

The development environment specifications:

Processor	Intel Core i5 or equivalent
RAM	8GB minimum
Storage	500GB HDD/SSD
Peripherals	Webcam, microphone

0.5.4 Results

The experimental results include the following interfaces:



Figure 4: Login Page

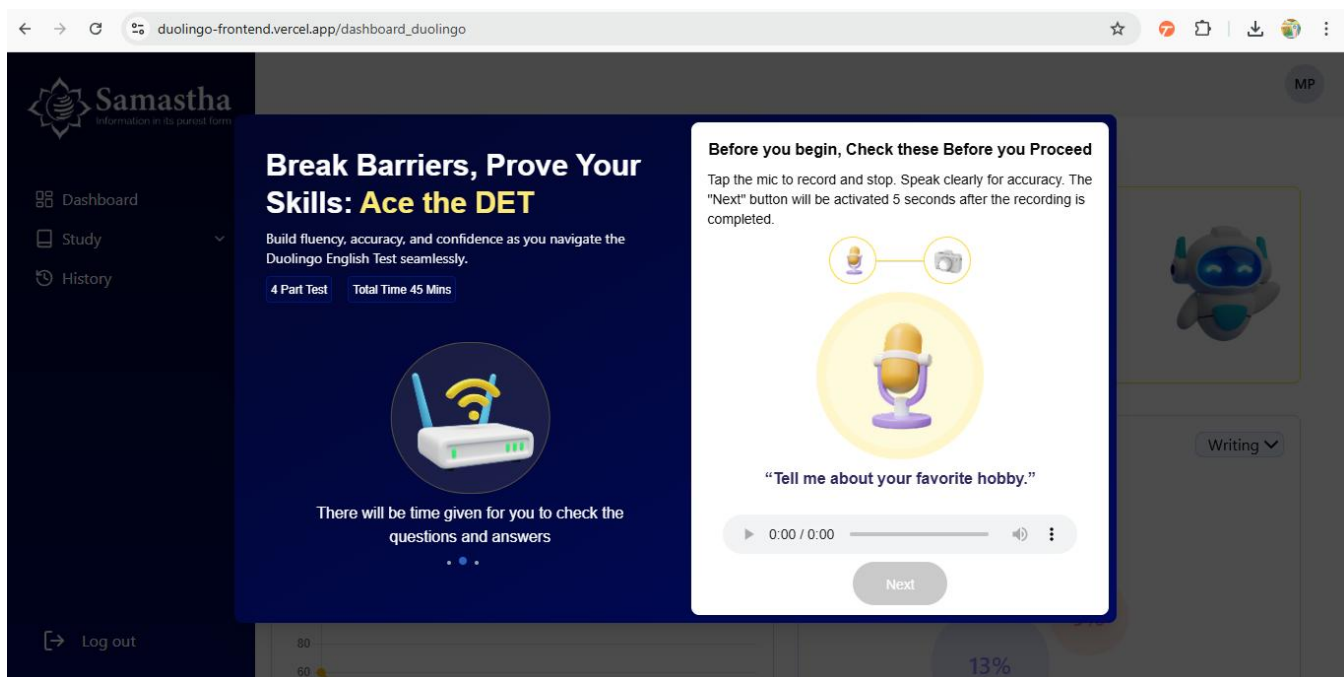


Figure 5: Device Check Page

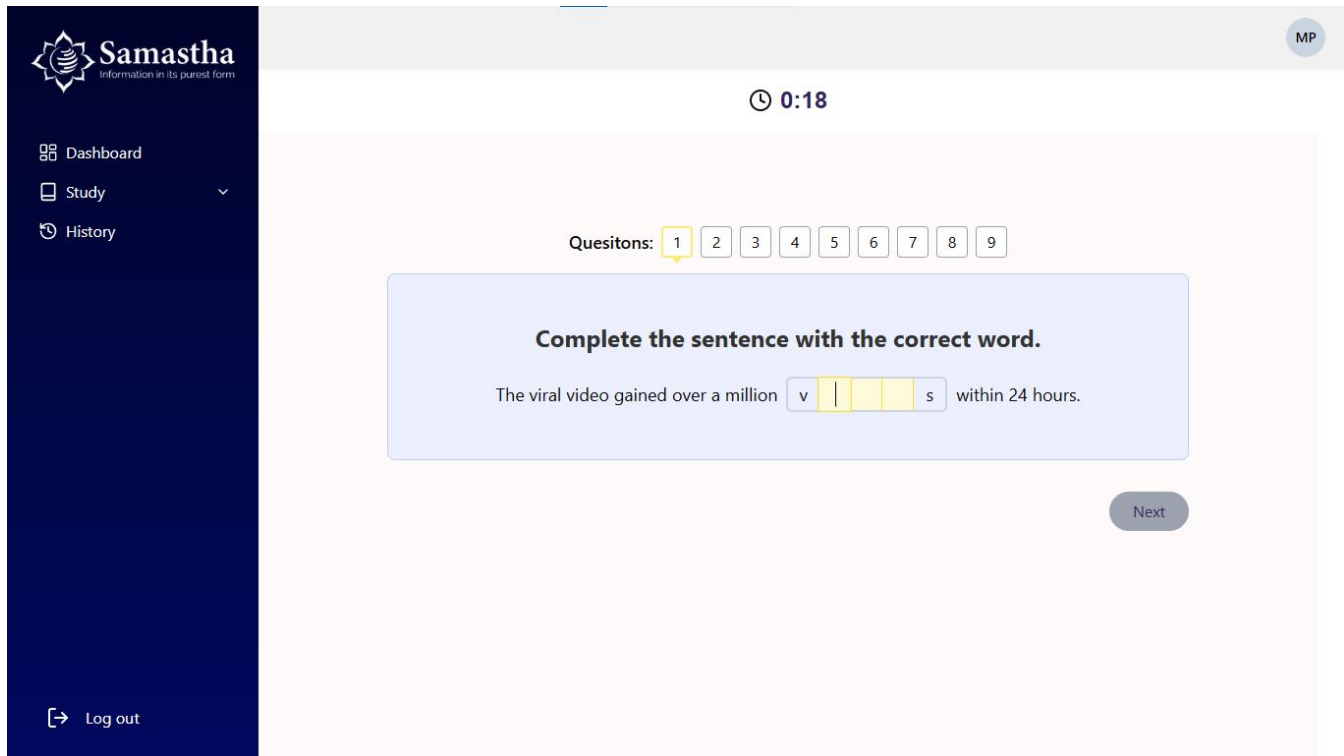


Figure 6: Test Interface

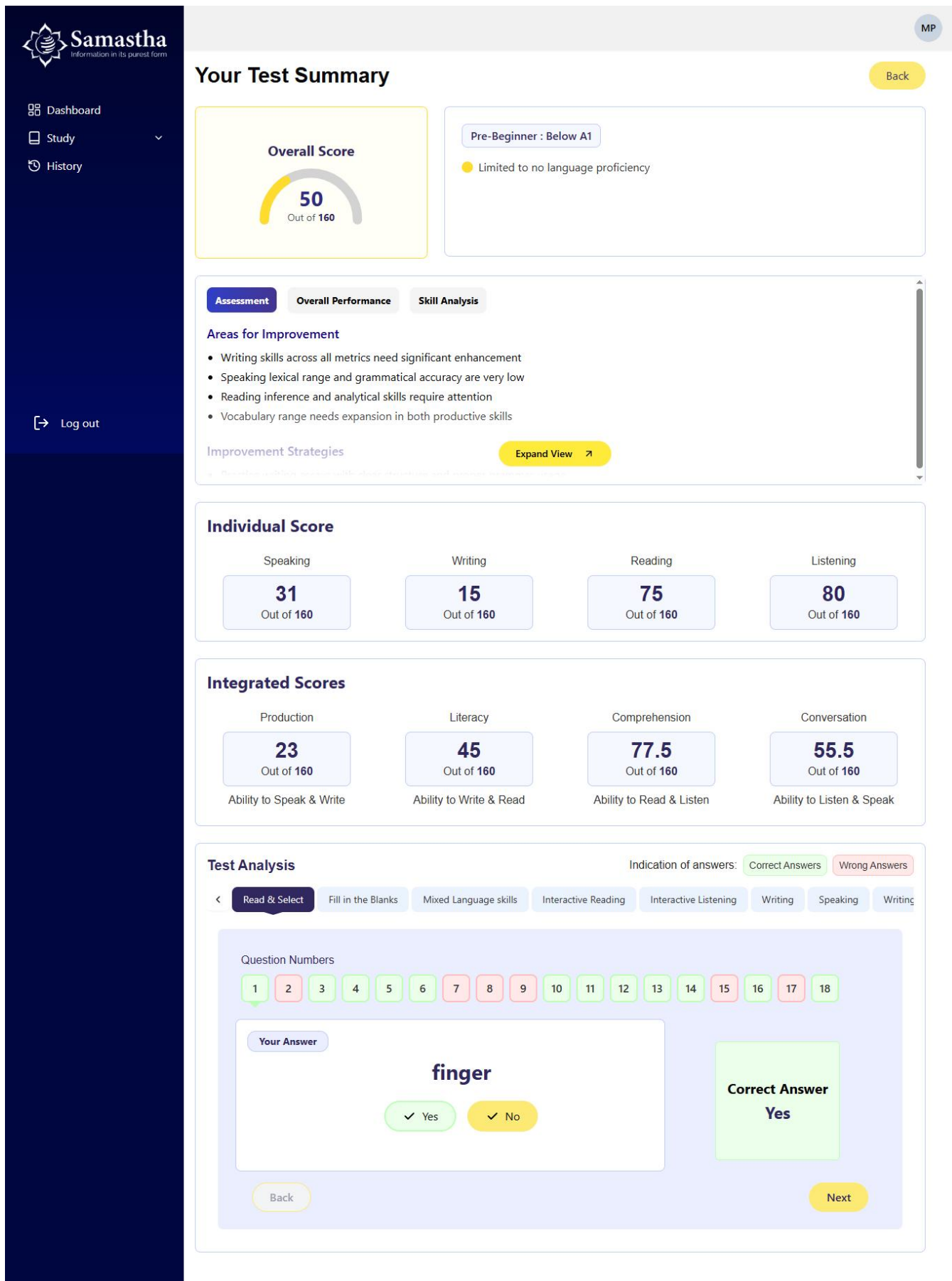
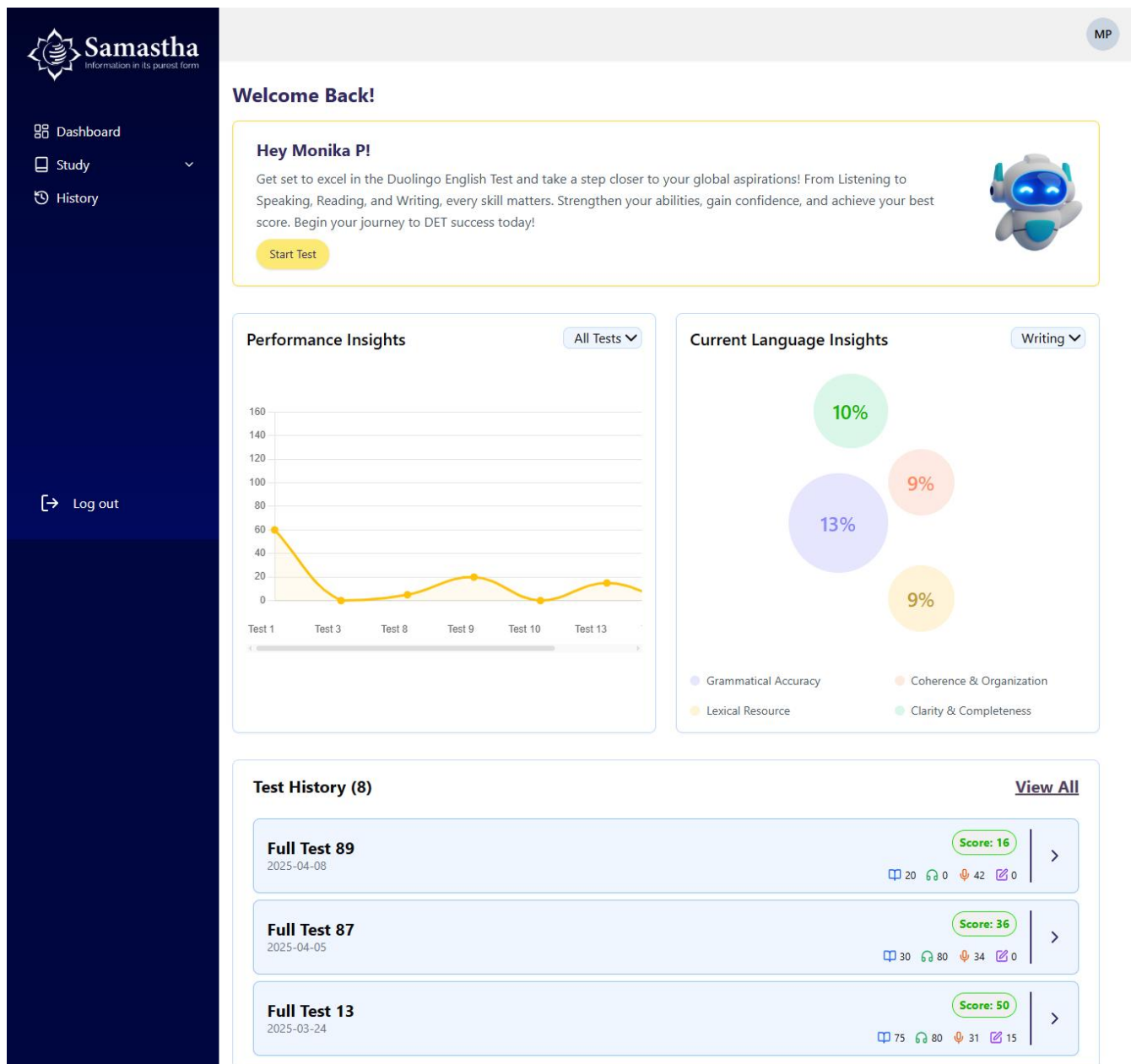


Figure 7: Feedback System



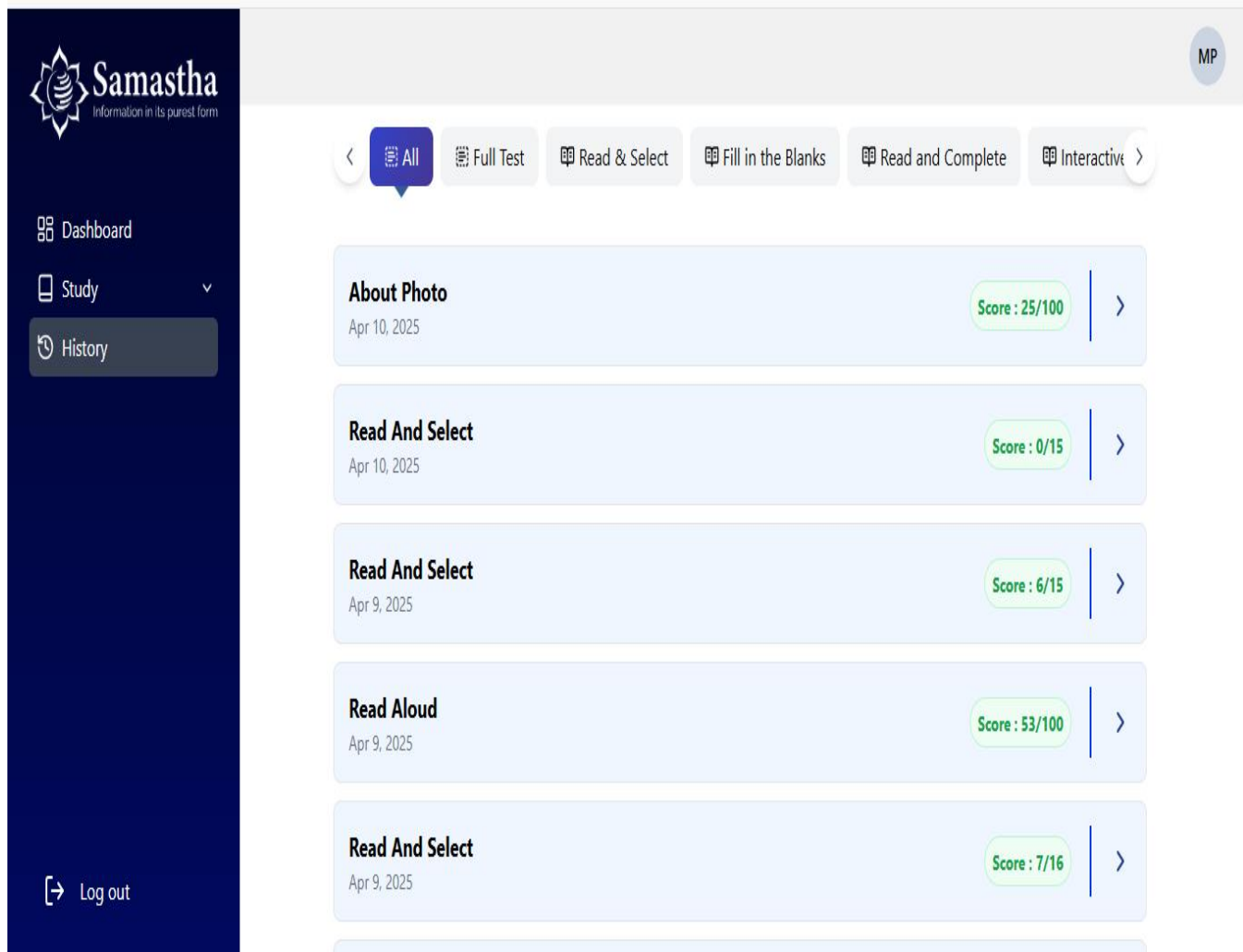


Figure 9: User History



Samastha
Information in its purest form

- Dashboard
- Study
- History

Log out

MP

Your Proficiency Level

Grammar Accuracy
Coherence
Vocabulary Usage
Clarity Completeness

Dashboard

About the Metrics

Evaluates your ability to use correct grammar, sentence structures, punctuation, and spelling in writing. Fundamental for creating professional, credible written communication in academic and workplace settings.

7%

Grammar Accuracy

Overview of your Level

Your grammatical accuracy currently shows beginner-level proficiency with a 7% overall average, indicating fundamental challenges with basic grammatical structures. While there has been some improvement in Write-About-Photo tasks since March 24th, recent test scores (particularly Tests 89 and 87 at 0%) suggest a need for focused attention on core grammar concepts.

Areas to Focus

What Went Well

- Building on your previous Interactive-Writing challenges, concentrate on basic subject-verb agreement and present tense conjugations, as these fundamental skills will help improve your consistently low test scores
- Given your continued struggles with Writing-Sample tasks, focus on proper article usage (a/an/the) and plural formations, which appear as persistent areas needing development across multiple assessments



Samastha
Information in its purest form

- Dashboard
- Study
- History

Log out

MP

Compare your Performance

Last 3 Tests

Select any two tests to review and compare your performance.

Practice test: 13

Band score
15
Grammar Accuracy
20%

Key Weakness
No Key Weakness

Key Strength
No Key Strength

Question Types
Write About Photo, Writing Sample, Interactive Writing

View Feedback

Practice test: 87

Band score
0
Grammar Accuracy
0%

Key Weakness
No Key Weakness

Key Strength
No Key Strength

Question Types
Write About Photo, Writing Sample, Interactive Writing

View Feedback

Your Performance

Total Tests: 3



Figure 10: Proficiency Page

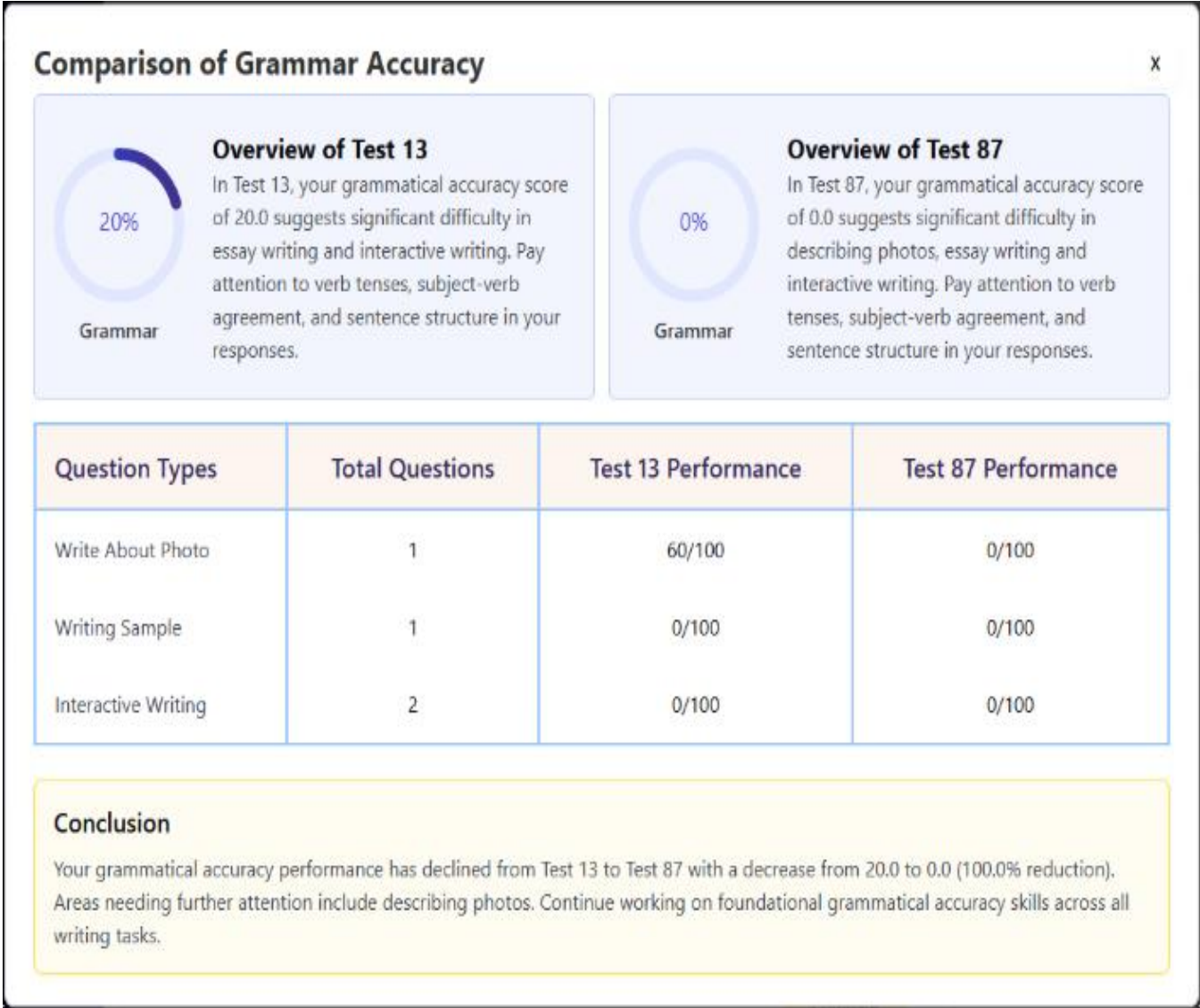


Figure 11: Comparison Page

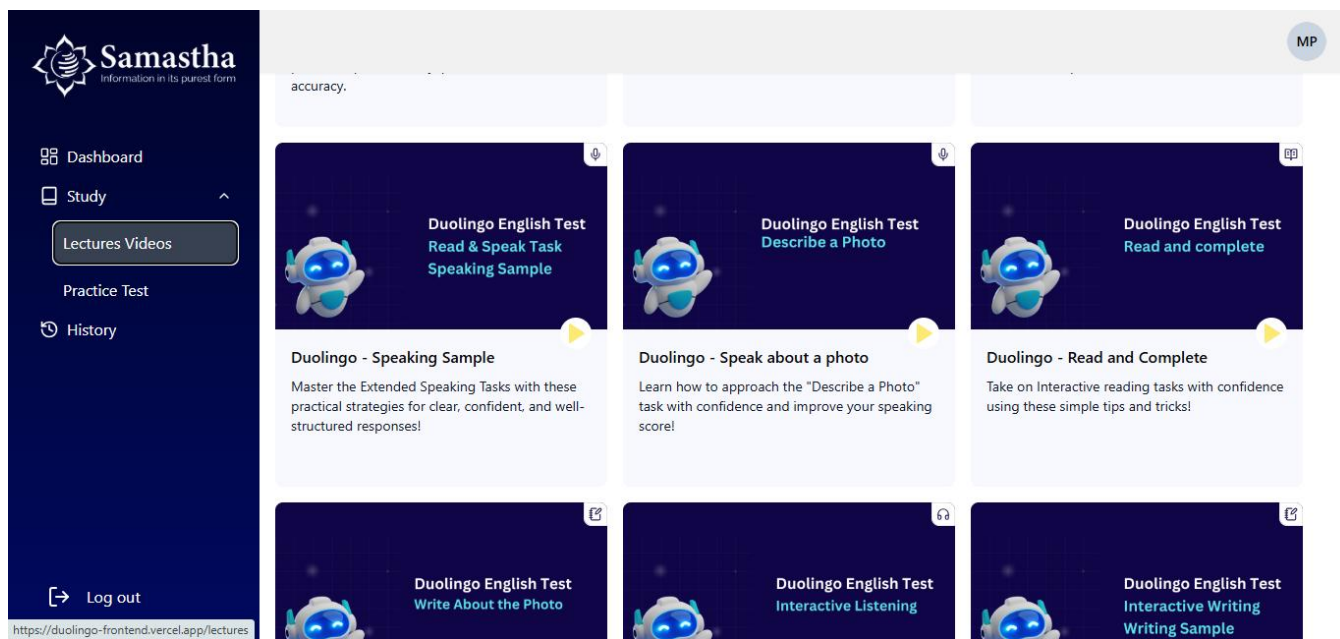


Figure 12: Study Page

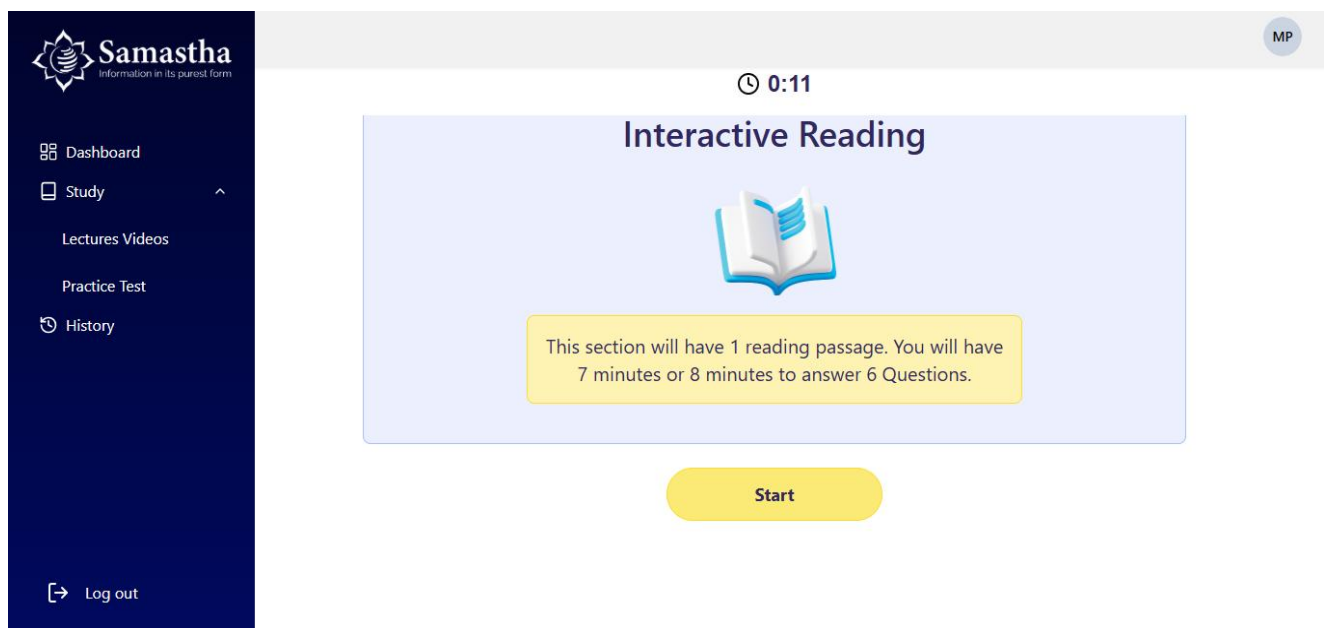


Figure 13: Practice Test

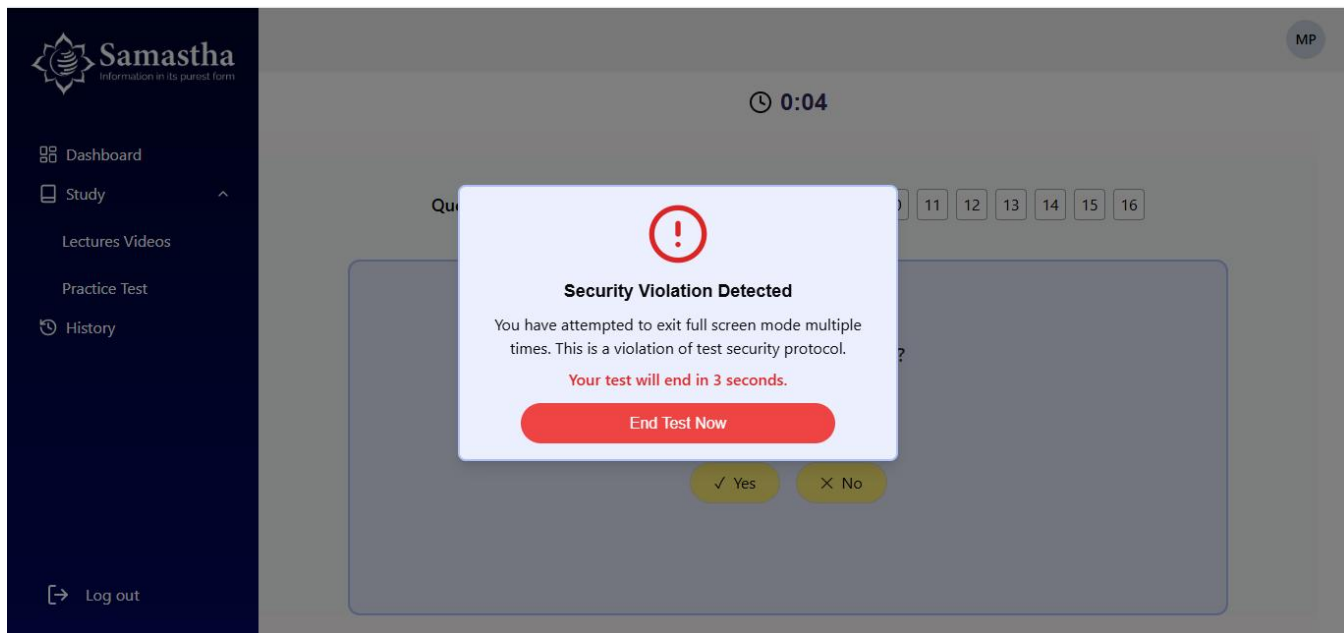


Figure 14: Warning Page

0.5.5 Results Comparison

Performance metrics comparison:

Average Writing improvement	+14%
Average Speaking improvement	+17%
User satisfaction rate	92%
System accuracy	89%

0.6 Conclusion

The Duolingo Mock English Test System successfully simulates the structure, experience, and evaluation process of the official Duolingo English Test (DET), offering users a comprehensive, intelligent, and personalized platform for English proficiency practice [3]. Through a full-stack web implementation using Flask and React.js, and integration with Azure Cognitive Services, the system ensures real-time response evaluation, adaptive question delivery, and dynamic feedback across all four language skills [5][9]. By incorporating AI-driven scoring models, the system provides learners with detailed insights into their language abilities, mapped to CEFR standards [7][8]. Features such as adaptive difficulty adjustment, voice recognition, task-specific feedback, and progress dashboards contribute to a more engaging and effective learning experience [4][6]. This project addresses key limitations of traditional language preparation methods by enabling learners to prepare on their own time, from anywhere, with intelligent feedback that targets their specific weaknesses and tracks measurable improvement [6].

0.7 Future Scope / Directions

The system establishes a foundation for several enhancements, including expanded feature sets, broader platform support, advanced analytics, and academic integration [9]. Our grammar and vocabulary modules have been completely redesigned with specialized practice modes that adapt to individual learning styles [4]. Each module now offers targeted exercises that reinforce specific language skills through spaced repetition and contextual learning **spaced`repetition** [8]. Complementing these modules is a new real-time hint system, providing AI-generated guidance precisely when needed, ensuring learners never feel stuck without compromising the learning process. Additionally, the newly introduced voice feedback playback feature enables users to record and review their speech, comparing their pronunciation against native speaker models to perfect accent and intonation **speech`feedback** [5].

Platform accessibility has also been significantly expanded. Mobile applications for both iOS and Android have been launched, bringing the full learning experience to smartphones and tablets **mobile`learning** [9]. These applications support offline functionality with deferred synchronization, allowing learners to continue their studies even without internet access; progress automatically updates upon reconnection [9]. Furthermore, the multilingual interface now supports over 20 languages, broadening accessibility to a global audience **multilingual`interface** [7].

Significant advancements have been made in artificial intelligence and analytics. Our cutting-edge emotion and stress detection technology analyzes vocal patterns to identify levels of confidence or struggle, dynamically adjusting lesson difficulty to better support the learner **emotion`detection** [8]. Personalized learning paths, powered by machine learning algorithms, analyze user performance and create customized curricula tailored to individual strengths and weaknesses **adaptive`learning** [4]. A new benchmarking system provides comparative insights by analyzing a learner's progress relative to peers with similar learning goals and backgrounds, offering valuable context and motivation **benchmarking** [6].

In terms of academic integration, a comprehensive teacher dashboard has been developed, offering institutional users unprecedented visibility into student progress [9]. Educators can now track detailed performance metrics across classes and identify specific areas requiring additional support **teacher dashboard** [6]. Students are able to earn certificates of completion upon achieving proficiency milestones, providing tangible recognition of their accomplishments **certificate system** [7]. Furthermore, a curriculum mapping feature ensures that all content is aligned with established academic standards across multiple educational frameworks, facilitating seamless integration into existing language programs **curriculum mapping** [7].

Looking ahead, integration with emerging technologies such as augmented reality (AR) and conversational AI presents transformative potential for immersive language learning experiences. AR-driven vocabulary drills and real-world situational simulations can enhance contextual understanding and learner engagement, particularly in speaking and listening modules. Furthermore, integration with wearable devices and voice assistants could enable passive language acquisition and continuous feedback outside traditional study settings.



Figure15 :Future work

0.8 References

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